TUM and Integral Polytopes

only a property of the matrix.

$\underbrace{\{x : Ax \le b\}}$ ~~integral~~ $\forall b$

For a fixed A, b, guarantee integrality.

max: $c^T x$

st: $Ax \le b$

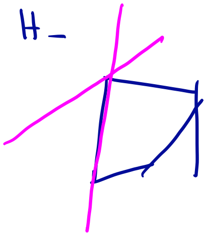
Faces and Supporting Hyperplanes — 1/3

$P = \{Ax = b\}$

$H = \{x : s^T x = b\}$ hyperplane.

H is called a supporting hyperplane

if $H \cap P \neq \emptyset$ and if $P \subseteq H_+, H_-$

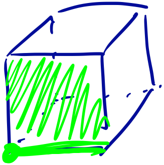


$F \subseteq P$ is called a

Face of P if

$F = \{x \in P : A'x = b'\}$, A', b' a subset of rows of (A, b)

Faces and Supporting Hyperplanes — 2/3



F is a minimal (inclusion-wise) face of P iff F is an affine subspace

i.e. $F = \{x \in \mathbb{R}^n : A'x = b'\}$.

Faces and Supporting Hyperplanes — 3/3

Thm: $P = \{Ax \le b\}$ $^{(A, b \text{ rat'l})}$ is integral iff each rational supp hyperplane contains integral vector.

pf: "$\Rightarrow$" obvious.

"$\Leftarrow$" wlog A, b integral. Let $F = \{x : A'x = b'\}$ minimal face.

Claim: If F contains no integral vector, then $\exists\, y > 0$ s.t. $y^T A'$ is integral, but $y^T b'$ is not. // Exercise.

Let $c \triangleq y^T A'$, $\gamma = y^T b'$ (non-integral).

$H = \{x : c^T x = \gamma\}$.

Exercise: $F = P \cap H$

c integral. H is supp. hyperplane.

$\Rightarrow$ H contains no integral vectors.